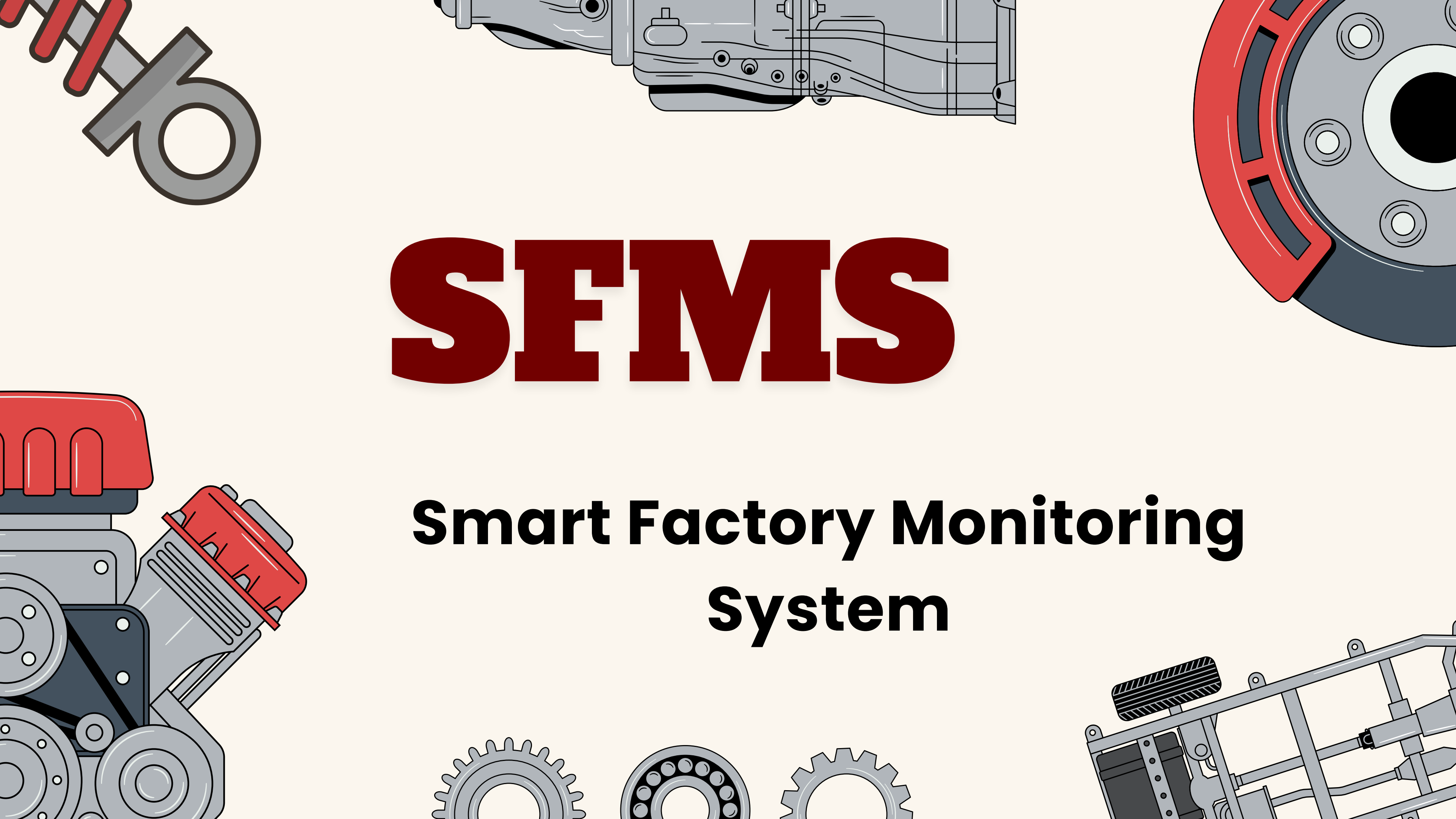


The background features several stylized mechanical components. In the top left is a red and grey bolt. In the top center is a grey mechanical arm or bracket. In the top right is a grey circular flange with a red ring. In the bottom left is a red and grey engine or motor. In the bottom center are three grey gears. In the bottom right is a grey mechanical assembly with a black roller.

SEMS

Smart Factory Monitoring System



Our Team

Basmala Moataz Mohamed

Hamss Abd Elgawad Mohamed Emara

Malak Ahmed Abo Elsood

Sara Muhammed Abdelmonem Younis

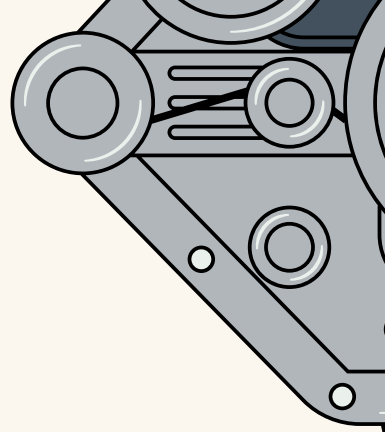
Project Motivation

Problem:

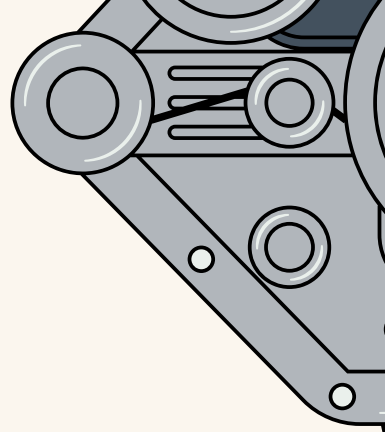
Unexpected machine failures cause production downtime.
Manual monitoring cannot detect issues early.
Reactive maintenance increases repair cost and production loss.

Our Solution:

Real-time monitoring of machine conditions using IoT sensors.
Continuous Health Score evaluation for every machine.
Early fault detection using Predictive Maintenance.
Instant alerts and automated maintenance actions.



Sensor Simulation & MQTT Data Layer



Simulate a smart factory and publish real-time sensor data

Implementation

- 4 Virtual Machines
- 4 Sensors per Machine
 - Temperature
 - Vibration
 - Current
 - RPM
- ESP32 Simulator (Wokwi)
 - Machine 2 uses real potentiometersTemperature Control
 - Vibration Control

Communication

ESP32



MQTT Publish



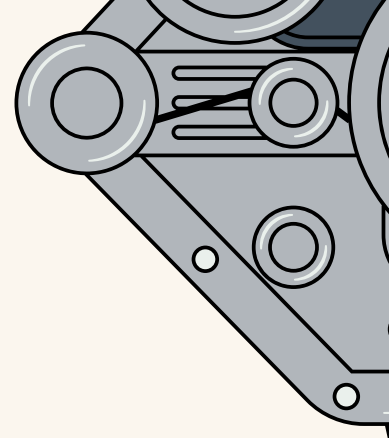
HiveMQ Cloud



Other Modules



Sensor Simulation & MQTT Data Layer



Simulate a smart factory and publish real-time sensor data

MQTT Topic Structure

**factory/
machine1/
temperature
vibration
current
rpm**

JSON Payload

```
{  
  "machine":"machine2",  
  "sensor":"temperature",  
  "value":72  
}
```



Demo Scenario

Machine 2 gradually:

↑ Temperature

↑ Vibration



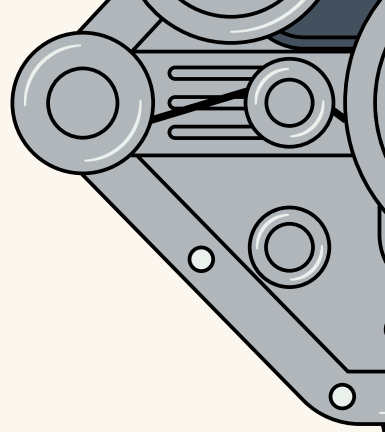
Health decreases

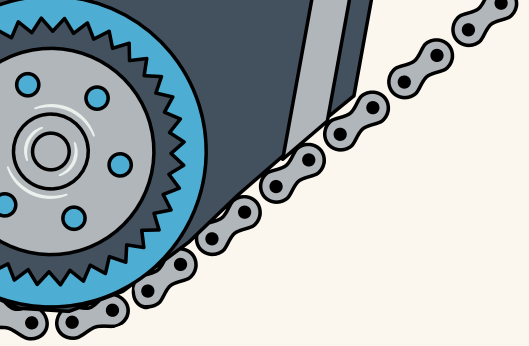


Failure Detection



Predictive Maintenance Trigger

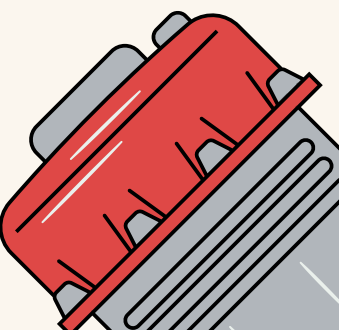
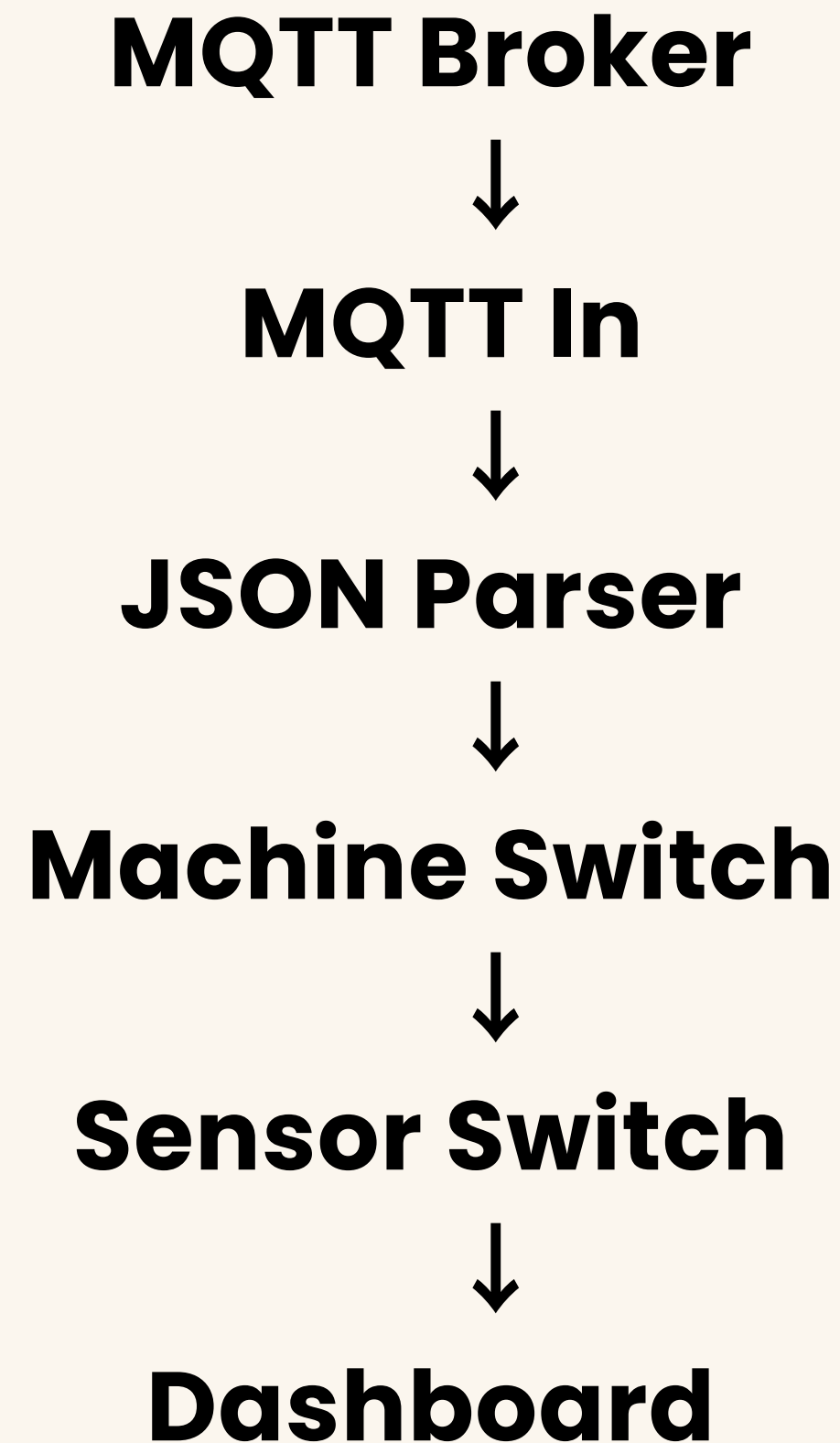




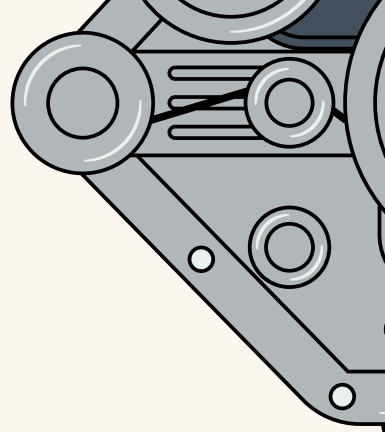
Node-RED Data Routing & Dashboard



MQTT Data Flow



Node-RED Data Routing & Dashboard



Machine Routing

Routes data to:

Machine 1

Machine 2

Machine 3

Machine 4

Sensor Filtering

Each machine is divided into

-Temperature

-Vibration

-Current

-RPM

Using

Switch Nodes

Change Nodes

Dashboard

Live

Gauges

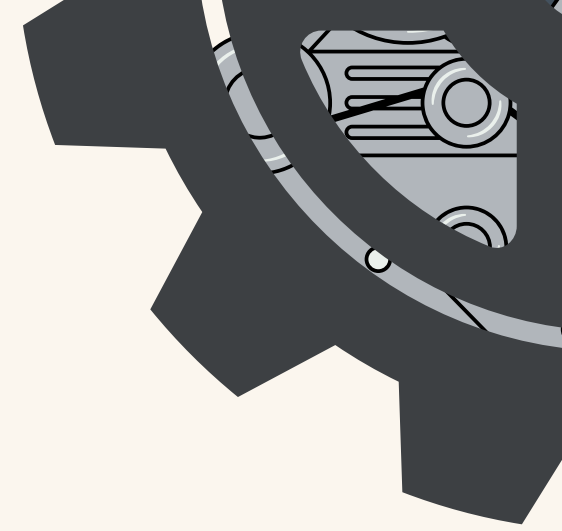
Charts

Real-Time

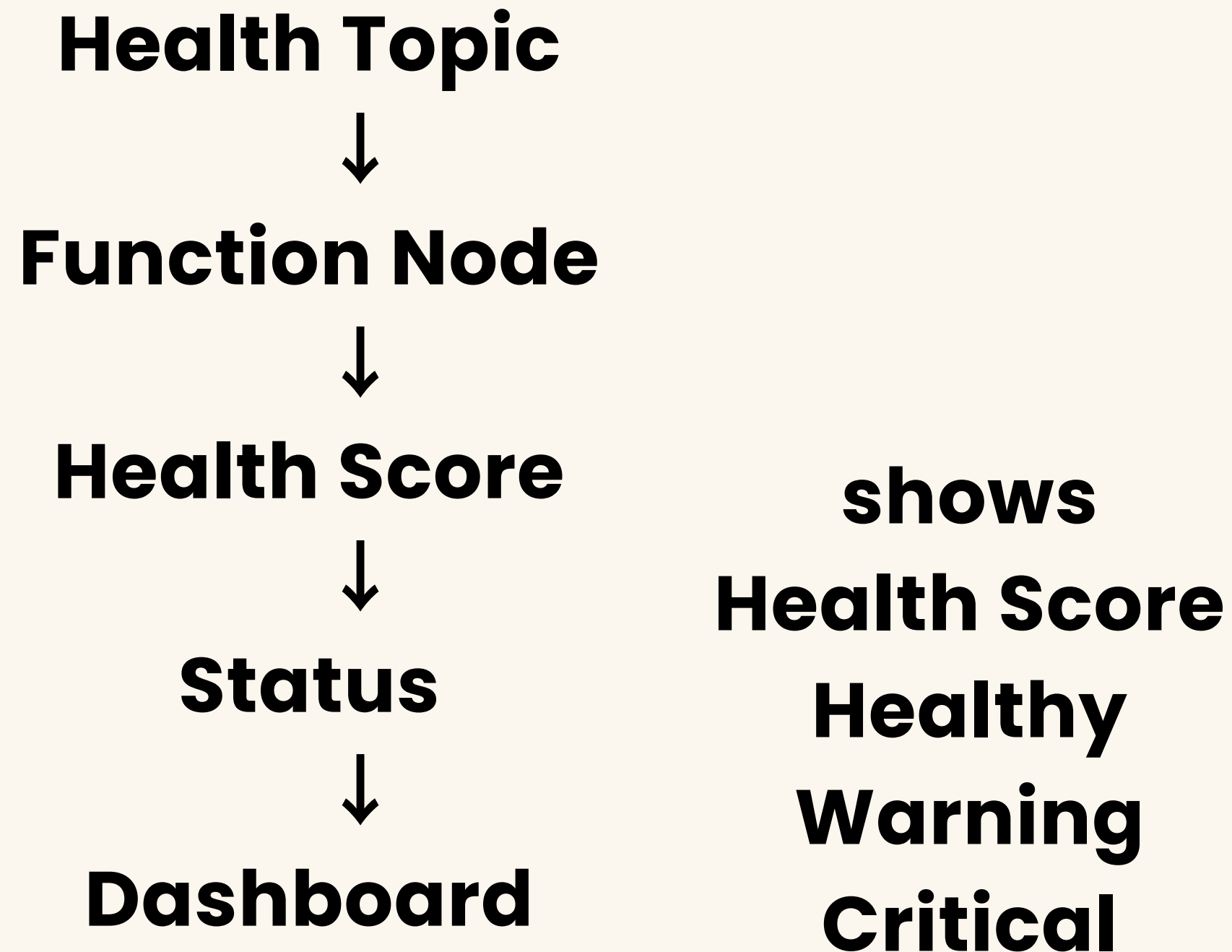
Values



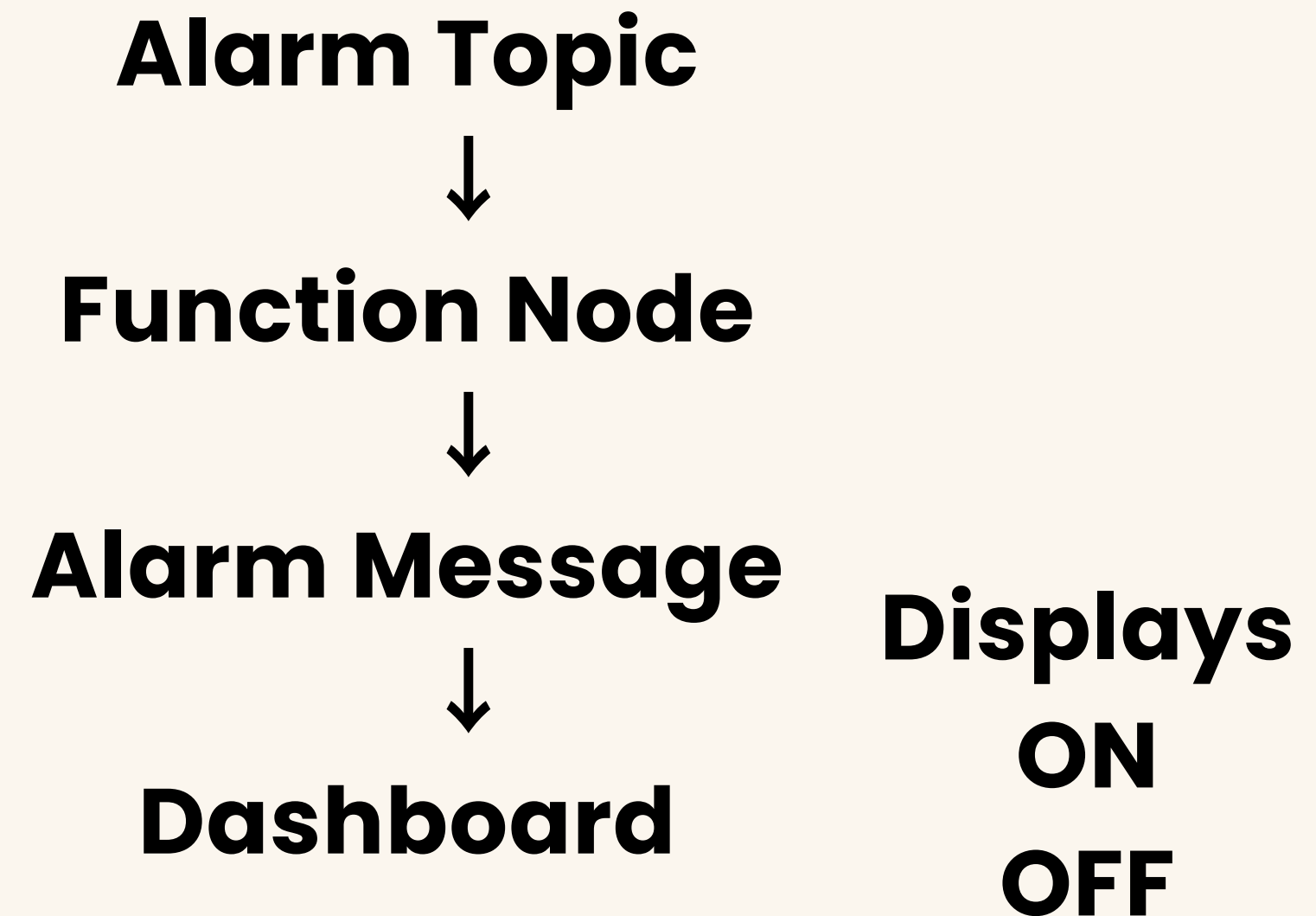
Node-RED Data Routing & Dashboard

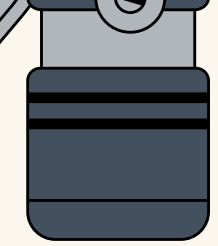


Health Status Monitoring

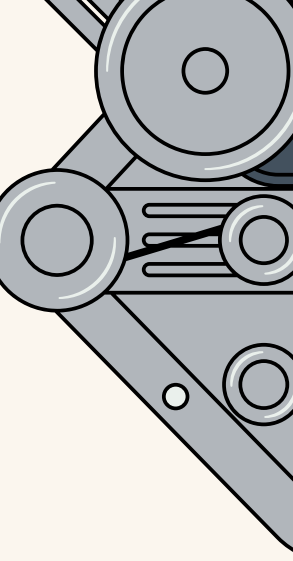


Alarm Monitoring





Health Score Calculation



Evaluate machine condition using sensor readings.

MQTT Sensors



Store Readings



Wait for 4 Sensors

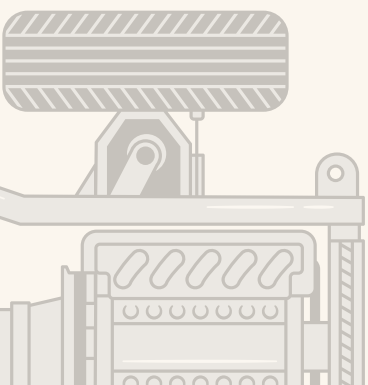


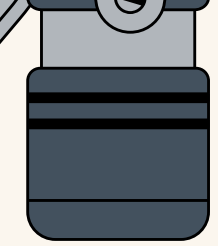
Calculate Health Score



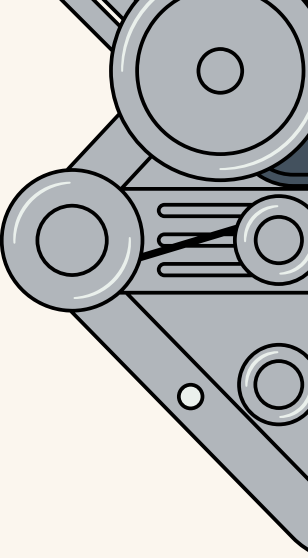
Publish Result

Inputs
Temperature
Vibration
Current
RPM





Health Score Calculation



Evaluate machine condition using sensor readings.

Health Score Rules

Temperature > 70°C



-20

Vibration > 5



-25

Machine Status

Score Status

80–100 Healthy

60–79 Warning

< 60 Critical

خارج RPM
المدى

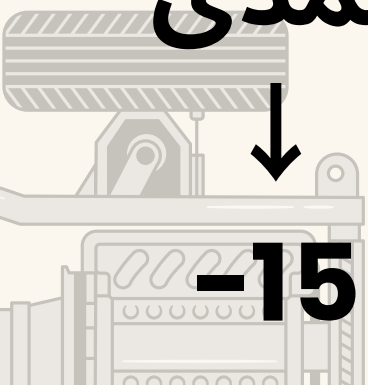


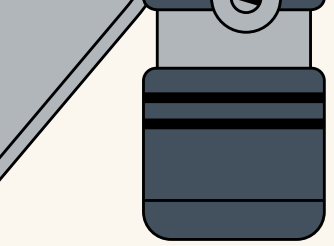
-15

Current > 15A

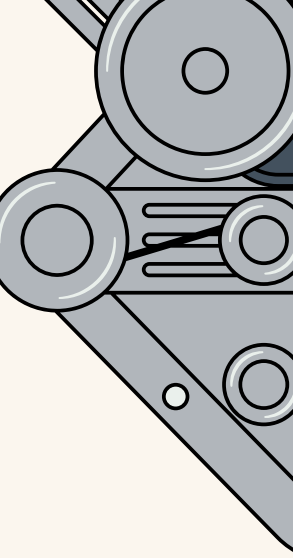


-20





Health Score Calculation



Evaluate machine condition using sensor readings.

Diagnosis

Output JSON

Possible outputs

Normal

Overheating

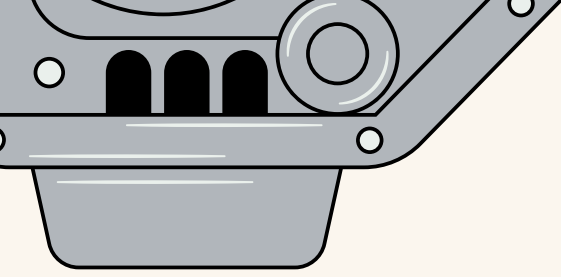
High Vibration

Motor Overload

Bearing Failure

```
{  
  "machine": "machine2",  
  "health_score": 55,  
  "status": "Critical",  
  "diagnosis": "Bearing Failure"  
}
```





Predictive Maintenance System



Final Decision Layer

**Receives
Health Score Results**



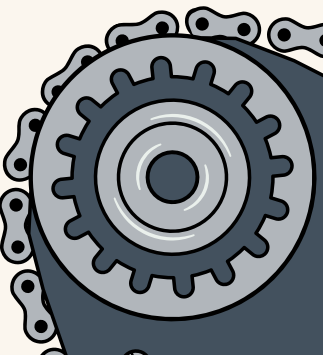
Analyzes Machine Condition

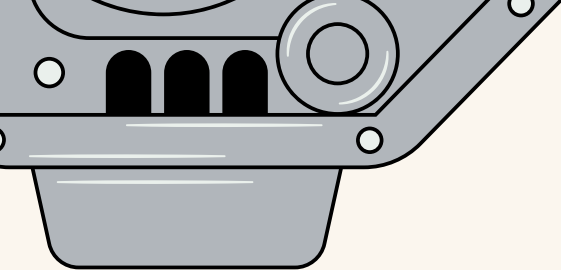


Determines Maintenance Action

Main Functions

**Store Machine History (CSV)
Generate Maintenance Ticket
Publish Alarm Status
Send Telegram Notification**





Predictive Maintenance System



Maintenance Ticket

Contains

- Ticket ID
- Machine Name
- Timestamp
- Health Score
- Diagnosis
- Priority
- Suggested Action



Notification Flow

Health Result



Critical?



Yes



Telegram Alert



Maintenance Team

Alarm Flow

Critical Machine



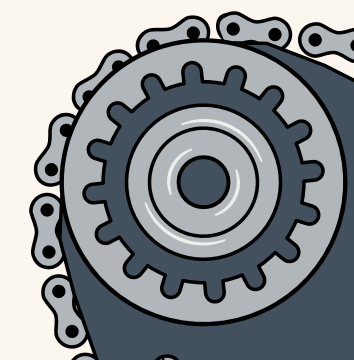
Alarm ON

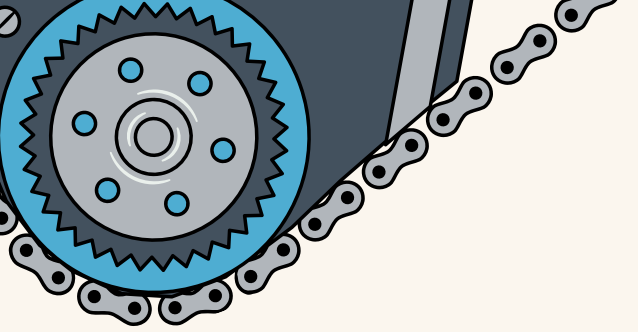


MQTT



Dashboard Indicator





Reliability Analytics

**Real-Time Analytics Module
Calculates**

- MTBF
- MTTR
- RUL

Metrics

MTBF

Average time between failures

MTTR

Average repair duration

RUL

Estimated remaining operating life

Health Score



Analytics

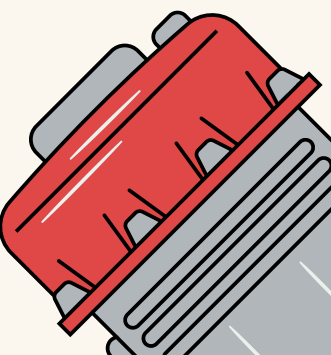


MQTT



Dashboard

Data Flow>>





Final Architecture

ESP32 Sensors



MQTT Broker



Node-RED Dashboard



Health Score



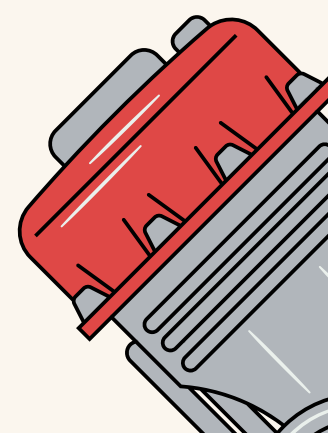
Predictive Maintenance



Analytics



Telegram + Dashboard + CSV



The background features several stylized mechanical components. In the top left is a grey bolt with red washers. In the top center is a grey mechanical arm or bracket. In the top right is a grey circular flange with a red ring. In the bottom left is a red and grey engine component. In the bottom center are three grey gears of different sizes. In the bottom right is a grey mechanical assembly with a black roller.

**Thank
You**